The Negative Impact of Technology in the K-12 Classroom: A Comprehensive Research Report

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Executive Summary

The evidence on the negative impact of technology in K-12 classrooms has reached a critical mass. Dr. Jared Cooney Horvath's meta-synthesis of 398 meta-analyses covering more than 21,000 studies found an overall weighted mean effect size of d = 0.29 for educational technology — below the d = 0.40 threshold that John Hattie identifies as the minimum for meaningful educational impact. International assessment data from PISA and TIMSS consistently show that heavy technology users score significantly lower than moderate or non-users. A growing body of experimental evidence — including a 2025 randomized controlled trial of approximately 17,000 students — demonstrates that removing phones from classrooms produces measurable academic gains, particularly for low-performing students.

At the same time, the counterarguments deserve serious attention. Oxford researchers Amy Orben and Andrew Przybylski have demonstrated that technology use explains at most 0.4% of the variation in adolescent well-being, and a 2024 meta-analysis in *Computers & Education* found that well-implemented educational technology does produce small but statistically significant positive effects on student achievement, with particularly meaningful effects for disadvantaged students. The research base claiming sweeping harm has methodological weaknesses, including reliance on correlational designs and publication bias.

Tennessee has emerged as a national leader in addressing these concerns. Governor Lee signed HB 932 (a statewide classroom cell phone ban) in March 2025, the Protecting Children from Social Media Act in 2024, and the Teen Social Media and Internet Safety Act in 2025. Early results from Tennessee districts implementing phone-free policies are promising: Emerald Academy in Knoxville reported that 80-90% of pre-ban discipline incidents were phone-related, KIPP Nashville saw a 37% decrease in chronic absenteeism, and East Nashville Magnet High School saw increases in students passing end-of-course exams.

The bottom line for a district like Greeneville City Schools: the cell phone ban is a well-supported first step, but the harder and more consequential question — how to manage school-issued devices (Chromebooks, laptops) for maximum instructional benefit and

minimum distraction — remains largely unaddressed by state or federal policy and will require local leadership, intentional pedagogy, and ongoing evaluation.

Key numbers:

- U.S. K-12 schools spend approximately $30 billion annually on education technology
- 67% of educational software licenses go unused, wasting more than $1 billion per year
- Students using computers 6+ hours daily score 66 PISA points lower than nonusers (equivalent to dropping from the 50th to the 24th percentile)
- 56% of educators identify off-task behavior on school-issued devices as a major distraction — ranking it above personal cellphones (38%)
- 35 states and D.C. have enacted laws restricting student cellphone use as of late 2025
- Sweden has committed EUR 104 million to reintroduce physical textbooks after declining reading scores

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1. Definition and Background

The Researcher: Dr. Jared Cooney Horvath

No researcher named "Jason Horvath" was identified in the literature. The most prominent Horvath in this space is **Dr. Jared Cooney Horvath**, a neuroscientist who has conducted research at Harvard University, Harvard Medical School, and the University of Melbourne. He has worked with over 1,000 schools worldwide and is the author of *The Digital Delusion: How Classroom Technology Harms Our Kids' Learning* (2025) and a widely cited 2024 After Babel article titled "The EdTech Revolution Has Failed." In January 2026, Horvath testified before the U.S. Senate Committee on Commerce, Science, and Transportation, presenting his synthesis of the educational technology evidence base (Horvath, 2026).

Horvath's central argument is that the educational technology industry has invested billions in classroom devices, software, and digital curricula without producing commensurate improvements in student learning. His meta-synthesis of 398 meta-analyses — the most comprehensive quantitative review of the ed-tech evidence base to date — found that technology's impact on learning falls below the threshold for meaningful educational effect (Horvath, 2025).

The Broader Context

The question of technology's role in K-12 education has shifted dramatically since 2020. The pandemic accelerated 1:1 device adoption to near-universality in American schools, with ESSER funding enabling districts to purchase millions of Chromebooks, iPads, and laptops. Simultaneously, youth mental health declined sharply, smartphone and social

media use among adolescents reached saturation levels, and international test scores revealed troubling patterns linking heavy device use to lower academic achievement (UNESCO, 2023; OECD, 2023).

By 2025, a counter-movement had emerged. Countries including Sweden, Denmark, the Netherlands, France, and Australia enacted or expanded school device restrictions. In the United States, 35 states passed legislation restricting student cellphone use. The U.S. Surgeon General issued an advisory calling social media an "urgent public health issue" for youth (HHS, 2023). And researchers including Horvath, Jonathan Haidt (*The Anxious Generation*), Jean Twenge (*iGen*), and Larry Rosen (*iDisorder*) built increasingly visible cases that technology in classrooms was causing more harm than good.

Key Terms

- **1:1 program**: A deployment model providing one computing device per student
- **Ed-tech**: Educational technology products including devices, software, digital curricula, and platforms
- **Screen inferiority effect**: The documented tendency for reading comprehension to be lower on screens than on paper (Delgado et al., 2018)
- **Brain drain effect**: The phenomenon whereby the mere presence of a smartphone reduces available cognitive capacity, even when the phone is not in use (Ward et al., 2017)
- **Yondr pouch**: A lockable neoprene pouch used to secure student phones during the school day

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2. Evidence of Effectiveness (or Ineffectiveness)

Horvath's Meta-Synthesis: d = 0.29

Horvath's synthesis of 398 meta-analyses covering more than 21,000 individual studies represents the most comprehensive quantitative review of educational technology's impact on student achievement. The overall weighted mean effect size was **d = 0.29** — below the d = 0.40 threshold that Hattie identifies as the minimum for meaningful educational impact (Horvath, 2025; Hattie, 2009).

Subject-specific findings reveal that technology performs best in mathematics (d = 0.33) and writing quality (d = 0.32), modestly in literacy (d = 0.25), and poorly in the sciences (d = 0.18). The one area where technology shows strong effects is for students with specific learning needs (d = 0.61), suggesting that assistive technology and targeted digital interventions remain valuable for specific populations even as general-purpose classroom technology underperforms (Horvath, 2025).

Subject Area	Effect Size (d)	Meta-Analyses	Individual Studies
Mathematics	0.33	22	1,060

Subject Area	Effect Size (d)	Meta-Analyses	Individual Studies
Literacy	0.25	17	736
Sciences	0.18	6	391
Writing Quality	0.32	6	75
Specific Learning Needs	0.61	10	216
Overall Weighted Mean	**0.29**	**398**	**21,000+**

International Assessment Evidence

Large-scale international assessments provide some of the strongest evidence of a negative relationship between heavy technology use and academic performance:

- **PISA 2022**: Students using computers 6+ hours daily scored **66 points lower** than nonusers — equivalent to dropping from the 50th to the 24th percentile. The OECD concluded that "the negative association between ICT overuse and performance is significantly negative in most participating countries" (OECD, 2023).
- **TIMSS 2019**: Daily computer users scored **41 points lower in math** and **51 points lower in science** compared to infrequent users.
- **Gorjon and Oses (2023)**: Using PISA 2018 data from 22 OECD countries, researchers found an inverted U-shaped relationship between ICT use and performance. Moderate use helps; intensive use causes underperformance equivalent to roughly **half an academic year** in countries like Estonia, Finland, and Spain.

The Smartphone Effect

Experimental and quasi-experimental studies consistently show that removing phones from classrooms improves academic outcomes:

Study	Design	Sample	Effect
Beland & Murphy (2016)	Quasi-experimental	4 English cities	+0.07 SD average; low achievers gain 14.23% SD
Sungu et al. (2025)	Randomized controlled trial	~17,000 students	+0.086 SD; strongest for low-performing students
Figlio & Ozek (2025)	Quasi-experimental (NBER)	~200,000 students, Florida	+1.1 percentile points (Year 2); reduced absences

The Sungu et al. (2025) RCT is particularly significant as one of the first large-scale randomized experiments on phone bans, providing stronger causal evidence than prior correlational or quasi-experimental designs.

The “Brain Drain” Effect

Ward et al. (2017) demonstrated that the mere physical presence of a smartphone — even face-down, even powered off — significantly reduces working memory capacity and fluid intelligence. This “brain drain” effect has direct implications for classrooms where devices sit on every desk: even when students are not actively distracted, the cognitive cost of having technology within reach is measurable.

Screen vs. Paper Reading

Delgado et al. (2018) meta-analyzed 54 studies (n = 171,055) and confirmed that paper reading produces better comprehension than digital reading. This “screen inferiority effect” is moderated by text complexity, time pressure, and reader ability — with struggling readers disproportionately harmed by screen-based reading. Mueller and Oppenheimer (2014) further found that students taking notes by hand retain conceptual understanding better than those using laptops, even when laptop users take more extensive notes.

Digital Distraction in the Classroom

A 2025 systematic review of digital distractions in education found that 51.2% of students use digital devices 1 to 3 times per class for non-academic activities, including texting (21.9%), social networking (14.1%), and checking the time (41.2%). Ravizza et al. (2017) demonstrated that internet use during lectures predicted worse exam performance, even after controlling for intellectual ability and motivation (Springer, 2025).

Mental Health Evidence

The mental health dimension of classroom technology is intertwined with the broader screen time debate:

- Clinical depression **doubled** among U.S. 12-17 year-olds between 2011 and 2019 (Twenge, 2020)
- Adolescents aged 12-15 spending more than three hours daily on social media face **double the risk** of depression and anxiety symptoms (HHS, 2023)
- 77% of high school students report frequent social media use (CDC YRBS, 2023)
- 46% of U.S. teens have experienced cyberbullying (Pew Research, 2023)
- Up to 95% of youth ages 13-17 use social media, with more than a third using it “almost constantly” (HHS, 2023)

Important Caveats on Evidence Quality

The evidence base is not uniformly strong. Several critical limitations must be acknowledged:

- **Most studies are correlational**, not causal. Mental health issues could drive phone use (reverse causation), or both could be caused by a third factor (Orben & Przybylski, 2019).
- **Effect sizes for screen time on well-being are trivial.** Orben and Przybylski (2019) found that technology use explains at most **0.4% of the variation** in adolescent well-being — comparable to the effect of eating potatoes.
- **Publication bias is significant.** From the same datasets, researchers could produce "roughly 10,000 papers showing negative screen effects, 5,000 indicating no effect, and 4,000 demonstrating positive technology effects" depending on analytical specifications (Orben & Przybylski, 2019).
- **Two-thirds of the studies cited in Haidt's *The Anxious Generation* were published before 2010**, and only 22 contain data on either heavy social media use or serious mental health issues among adolescents (Platformer, 2024).
- **A 2024 *Nature* review** by Candice Odgers directly challenged Haidt's thesis: "The book's repeated suggestion that digital technologies are rewiring our children's brains and causing an epidemic of mental illness is not supported by science" (Odgers, 2024).

The distinction between Horvath's work (focused on academic achievement with large-scale meta-analytic evidence) and Haidt/Twenge's work (focused on mental health with more contested correlational evidence) is important. The academic achievement case against classroom technology rests on stronger methodological ground than the mental health case.

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3. Federal and State Policy Context

Federal Framework

ESSA Title IV-A: The Student Support and Academic Enrichment (SSAE) program is the primary federal technology funding mechanism. Congress built in a deliberate constraint: **no more than 15% of Title IV-A Effective Use of Technology funds** may be spent on purchasing technology infrastructure, including devices, equipment, and software. This cap reflects congressional recognition that hardware alone does not improve outcomes (U.S. Department of Education, n.d.).

IDEA and Assistive Technology: The Individuals with Disabilities Education Act mandates that IEP teams consider whether each student with a disability would benefit from assistive technology devices and services. This creates a **direct legal tension** with blanket device bans — any phone or device restriction policy must include carve-outs for students whose legal plans require technology access. A January 2026 GAO report documented persistent barriers to effective assistive technology deployment, including limited staff knowledge, insufficient training, and funding constraints (GAO, 2026).

COPPA and FERPA: The FTC finalized major amendments to the COPPA Rule on April 22, 2025, expanding definitions of personal information and enhancing consent requirements. The December 2024 PowerSchool breach — compromising data for over 10 million students — underscored the real-world consequences of inadequate data security in education technology (FTC, 2025).

Surgeon General's Advisory: In May 2023, Surgeon General Vivek Murthy declared social media use by children an "urgent public health issue." In June 2024, he called for warning labels on social media platforms. The advisory specifically recommended that schools ensure classroom learning and social time are phone-free (HHS, 2023).

Federal Legislation (Pending):

Bill	Status	Key Provisions
Kids Online Safety Act (KOSA)	Passed Senate 91-3 (July 2024); reintroduced May 2025	Duty of care for platforms serving ages 13-17
COPPA 2.0	Reintroduced March 2025	Extends protections to teens through age 17
Focus on Learning Act	Introduced 2025	Surgeon General study on cellphone effects in K-12
UNPLUGGED Act of 2025	Introduced 2025	Federal approach to student phone use

NTIA Inquiry (2025): The National Telecommunications and Information Administration launched an inquiry into whether federal subsidies are pushing schools toward more screens without evidence that it helps children learn, and is considering whether to tie screen-time accountability to school subsidies (NTIA, 2025).

Tennessee Policy Landscape

Tennessee has emerged as a **national leader** in addressing technology's negative impacts on K-12 students. The American Legislative Exchange Council (ALEC) recognized Tennessee as a "Policy Champion" for responsible technology use in schools.

HB 932 — Statewide Classroom Cell Phone Ban (Signed March 28, 2025): Requires all public school districts and public charter schools to adopt policies prohibiting student use of wireless communication devices during instructional time. Sponsored by Representative Rebecca Alexander (R-Jonesborough), who stated: "Let's get back to the basics of kids being kids." The bill passed with strong bipartisan support (80-10 in the House) and was implemented for the 2025-26 school year. Tennessee's school superintendents association requested the legislation, indicating statewide consensus among district leaders (WBIR, 2025).

HB 1891 — Protecting Children from Social Media Act (Signed May 2, 2024): Requires social media companies to verify user age and obtain parental consent before minors under 18 can create accounts. Passed with only two "no" votes in the House and zero in the Senate (WBIR, 2024).

HB 825 — Teen Social Media and Internet Safety Act (2025): Directed TDOE to develop guidance for LEAs on teaching responsible use of social media, internet, and AI tools in grades 6-12. Phase 1 (2025-26) banned student access to social media on school internet except for educational purposes. Phase 2 (2026-27) requires age-appropriate instruction on topics including screen time management, permanence of online content, mental health impacts, cyberbullying, and evaluating AI-generated information (WPLN, 2026).

HB 1886 — School Internet Filtering (2026 Session): Would further restrict students' access to certain online content and require annual compliance audits. Passed House Education Committee 18-1 (Nashville Banner, 2026).

Greeneville City Schools: Following the state mandate, the Greeneville City Board of Education approved its cell phone policy in June 2025, requiring students to turn off or silence personal electronic devices, store them out of sight, and not wear smartwatches or headphones during school hours. Exceptions exist for IEP/ILP requirements and assistive technology (WGRV, 2025).

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4. What Works: Implementation Models

Phone-Free Schools: The Evidence

Districts implementing phone-free policies have produced consistent, measurable results across diverse contexts:

District	Enrollment	Method	Key Result
Florida District (NBER study)	~200,000	All-day backpack storage	+1.1 percentile points in test scores (Year 2)
Richardson ISD, TX	~37,000	Yondr pouches	42% drop in out-of-school suspensions; 98% of teachers report fewer disciplinary issues
DeKalb County, GA	~93,000	Yondr + phone lockers	70% drop in due process referrals; 17% decrease in overall discipline
Cincinnati PS, OH	~33,000	Yondr pouches	12 high schools

District	Enrollment	Method	Key Result
			reduced 9th-grade failure rates; 93% of administrators observed improvement
Russellville, KY	~1,062	Yondr pouches	80% decrease in referrals
Walla Walla, WA	~5,600	Yondr pouches	Positive parent feedback; students report feeling safer
Pathway School, PA	Special ed	Full device ban	Transformed social engagement and lunchroom dynamics

Tennessee District Results

District	Action	Result
Emerald Academy, Knoxville	Yondr pouches (grades 6-8)	80-90% of pre-ban discipline was phone-related; improved social interactions, stronger peer relationships
KIPP Nashville	Locking device pouches	37% decrease in chronic absenteeism; attendance rose to 95%; higher test scores
East Nashville Magnet HS	Yondr pouches	83-87% of pre-ban referrals were phone-related; increase in students passing EOC exams

International Reversals

Country	Action	Key Result
Sweden	Committed EUR 104M to reintroduce textbooks; phone ban by 2026	Response to PIRLS reading decline (555 to 544 points, 2016-2021)
Netherlands	National phone ban, all school levels (2024-25)	75% of schools report improved concentration
Australia (NSW)	Public school phone ban	87% of students less distracted; 81% improved

Country	Action	Key Result
		learning
Australia (South Australia)	Public school phone ban	63% decline in social media incidents; 54% fewer behavioral issues
France	National phone ban (2018; expanded 2024)	Reduced cyberbullying; mixed evidence on grades alone
Norway	Directorate recommendations (2024)	96% of primary schools adopted bans within one year
Denmark	Smartphone/tablet ban law (2025-26)	Rolling back digital learning; returning to textbooks

Key Implementation Lessons

1. **Stricter policies produce better results.** RAND research confirmed: "The stricter the policy, the happier the teacher and the less likely students are to be using their phones." The Florida all-day policy outperformed the statewide classroom-only requirement.
2. **Physical enforcement mechanisms matter.** Yondr pouches and phone lockers outperform policies relying on teacher enforcement alone.
3. **Community engagement before rollout is critical.** Lebanon City Schools (OH) used community book studies of Haidt's *The Anxious Generation* to build buy-in before implementing restrictions.
4. **Discipline spikes are temporary.** Florida data shows suspensions climbed 12% in year one but returned to normal by year two.
5. **Equity in enforcement requires deliberate attention.** Without intentional planning, phone bans disproportionately increase discipline for Black students, at least in the short term (Hechinger Report, 2025).
6. **Small districts can succeed.** Russellville, KY (1,062 students) and Emerald Academy (Knoxville) demonstrate the model scales effectively with modest investment.

The Unresolved Problem: School-Issued Devices

A May 2025 EdWeek Research Center survey of 1,268 educators found that **56% identified off-task behavior on school-issued laptops, tablets, or desktops as a major distraction cutting into learning time** — ranking it above cellphones at 38%. One-third of educators reported that when students use school-provided devices, they are typically off-task more than 25% of the time. Tennessee's cell phone ban addresses personal devices, but school-issued 1:1 Chromebooks and laptops remain in student hands and are a documented source of equal or greater distraction (Education Week, 2025).

5. Cost-Benefit Analysis

The Scale of K-12 Technology Spending

American K-12 schools spent approximately **$30 billion on education technology in 2024**, a figure projected to nearly double by 2033. Per-pupil technology costs range from $142 to $490 per year in older estimates, though current 1:1 programs likely push this higher (Education Week, 2025; RAND).

The True Cost of 1:1 Programs

Cost Category	Chromebook	iPad
Device Cost (per unit)	$99 - $300	$449 - $768
Typical Refresh Cycle	4 - 6 years	4 - 5 years
Insurance/Replacement	$25 - $40/yr per device	$30 - $50/yr per device
Software Licensing	~$75/student/yr	~$75/student/yr

The PIRG "Chromebook Churn" report (2023) found that Google's built-in software expiration dates force premature device replacement. **Doubling the lifespan of Chromebooks could save taxpayers $1.8 billion** (PIRG, 2023).

The most infamous 1:1 failure — the Los Angeles Unified School District's **$1.3 billion iPad program** (2013) — collapsed due to unfinished software, security bypasses, insufficient infrastructure, and inadequate teacher training. The FBI investigated the procurement; Pearson paid only $6.45 million in settlement (NPR, 2014).

Software Waste

Research by Glimpse K12 found that **67% of educational software licenses go unused**, with some districts reaching 90%. Districts spend approximately **$3.8 billion per year** on ed-tech licenses, and more than **$1 billion is wasted annually** on software that fails to deliver value. Denver Public Schools reduced its digital tools from over 1,000 to 346, saving approximately $1 million from a $21 million annual ed-tech budget (Education Week, 2025; Glimpse K12, 2019).

Digital vs. Traditional Materials

Implementation costs for e-textbooks on iPad tablets are **552% higher** than new print textbooks in an average high school when infrastructure, devices, training, and recurring licensing are included (Britannica; Education Week, 2012).

Cybersecurity Costs

Metric	Value
K-12 ransomware attacks (2024-2025)	96 incidents

Metric	Value
Schools experiencing a cyber incident (July 2023 - Dec 2024)	82%
Average ransomware recovery cost	$3.76 million
Districts with no full-time cybersecurity staff	Two-thirds

The average ransomware recovery cost of $3.76 million is equivalent to approximately **54 teaching positions** (GovTech, 2025; Security Magazine).

Phone Ban Implementation Costs

District/System	Yondr Cost	Notes
Per student (typical)	$25 - $30	Per pouch
Cincinnati Public Schools	$500,000	Grades 7-12
New York City	$29 million	District-wide
Los Angeles Unified	$5.2 million	Largest U.S. district
Texas Phone Free Schools Grant	$20 million statewide	$12.50/student (HS), $10/student (MS)

The Financial Bottom Line

The financial case for re-evaluating classroom technology is substantial. Districts are spending billions on devices, software, and infrastructure while simultaneously facing waste from unused licenses (67%), costs from cybersecurity incidents ($3.76M average recovery), and new expenditures for phone bans and mental health services driven in part by the very technology they purchased. When technology's academic effect size (d = 0.29) falls below the threshold for meaningful impact, the return on investment is difficult to justify at current spending levels.

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6. Criticisms, Limitations, and Risks

Methodological Weaknesses in the Anti-Technology Evidence

The case against classroom technology, while growing, has significant evidentiary gaps:

Trivial well-being effect sizes. Orben and Przybylski's 2019 specification curve analysis across three large datasets (350,000+ adolescents) found that technology use explains at most **0.4% of the variation in adolescent well-being**. They demonstrated that "eating potatoes has nearly as negative an effect on adolescent mental health as screen use, and wearing glasses has a more negative effect" (Orben & Przybylski, 2019).

Candice Odgers' *Nature* rebuttal. A UC Irvine professor published a 2024 review directly challenging Haidt: "Hundreds of researchers, myself included, have searched for the kind

of large effects suggested by Haidt. Our efforts have produced a mix of no, small and mixed associations" (Odgers, 2024).

Pete Etchells' *Unlocked* (2024). Argues that the evidence for digital devices causing poor mental health, sleep deprivation, and shrinking attention spans is "weak" and that the current discourse represents "just the latest moral panic" in a long historical pattern that has included novels, radio, television, and video games (Etchells, 2024).

Researcher degrees of freedom. From identical datasets, researchers could have produced approximately 10,000 papers showing negative effects, 5,000 showing no effect, and 4,000 showing positive effects — depending on analytical choices (Orben & Przybylski, 2019).

Technology IS Effective Under Certain Conditions

A 2024 meta-analysis in *Computers & Education* (740 estimates from 72 experimental and quasi-experimental studies) found that educational technology initiatives have "a small, positive, statistically significant effect" on student achievement that "remains even after correcting for publication bias." For **less advantaged students**, the effect is moderate and meaningful (Di Pietro & Munoz, 2024).

Condition	Effect on Learning
Computer-assisted learning with feedback	Significant positive
Adaptive/personalized platforms	Positive, especially in STEM
Gamified/challenge-based learning	Strong positive (89.45% improvement over lecture)
Simple device access without pedagogy change	No significant effect
Technology replacing existing activities 1:1	No significant effect

Equity Concerns with Technology Reduction

The digital divide is real. 35% of low-income families with school-aged children lack home internet access. Black and Hispanic students are 1.3 times more likely than white peers to lack home broadband. For many students, school is the only place they interact with technology in meaningful, guided ways (Pew Research; Center for American Progress).

Workforce readiness. 92% of jobs require digital skills, yet one-third of workers have low or no digital skills. 55% of graduates said their programs did not prepare them to use generative AI tools (National Skills Coalition; Cengage, 2024).

Disability Rights and Medical Needs

Blanket technology restrictions create **legal conflicts with IDEA and Section 504**. Students with Type 1 diabetes using continuous glucose monitors (CGMs) need

smartphone connectivity for real-time blood sugar monitoring. Parents in 40 states have reported trouble getting CGMs monitored in schools, and parents in 10 states have filed federal complaints. The Supreme Court's 2025 decision in *A.J.T. v. Osseo Area Schools* made it easier for families to sue for damages when schools confiscate devices needed for medical monitoring (Healthbeat, 2025; AdvaMed, 2025).

Parent Concerns

A National Parents Union survey found 78% of parents want their children to have cellphone access during the school day in case of emergency. In the context of school shootings, 58% cited the need to reach their children, and 48% cited transportation coordination (NPU, 2024).

Conflicting Findings Flagged

The following contradictions exist within the evidence base:

1. **Academic impact**: Horvath's meta-synthesis finds d = 0.29 (below meaningful threshold) while Di Pietro & Munoz's 2024 meta-analysis finds small but significant positive effects. The difference may be explained by implementation quality — Horvath's broader synthesis includes many poorly implemented programs.
2. **Mental health**: Haidt and Twenge report dramatic increases in depression correlated with smartphone adoption, while Orben, Przybylski, and Odgers find trivial effect sizes from the same era. The disagreement centers on methodology (population-level trends vs. individual-level specification curve analysis).
3. **Phone bans and mental well-being**: One study comparing 30 schools with restrictive versus permissive policies found **no significant differences** in student mental wellbeing, anxiety, depression, or sleep — even as academic and behavioral outcomes improved. This suggests phone bans may improve classroom functioning without directly addressing the mental health concerns that motivate them.

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7. Measurable Outcomes

Academic Achievement

Outcome	Evidence Strength	Direction	Key Citation
Test score gains from phone bans	Strong (RCT + quasi-experimental)	Positive (0.086 SD; 1.1 percentile)	Sungu et al., 2025; Figlio & Ozek, 2025
Reading comprehension (paper vs. screen)	Strong (meta-analysis, n=171,055)	Paper superior	Delgado et al., 2018
Overall ed-tech effect on achievement	Strong (398 meta-analyses)	Below meaningful threshold (d=0.29)	Horvath, 2025

Outcome	Evidence Strength	Direction	Key Citation
Intensive ICT use vs. moderate use	Strong (PISA, 22 OECD countries)	Negative (~0.5 academic year loss)	Gorjon & Oses, 2023

Behavioral Outcomes

Outcome	Evidence Strength	Direction	Key Citation
Discipline referrals post-phone ban	Moderate (multiple districts)	42-80% reduction	Richardson ISD; Russellville; DeKalb County
Chronic absenteeism post-phone ban	Moderate (district-level)	37% decrease	KIPP Nashville
Student social interaction	Moderate (observational)	Increased	Multiple districts
Cyberbullying incidents	Moderate (district-level)	63% decline (South Australia)	Education South Australia, 2024

Mental Health

Outcome	Evidence Strength	Direction	Key Citation
Depression rates among teens (2011-2019)	Strong (epidemiological)	Doubled	Twenge, 2020
Social media and depression risk (3+ hrs/day)	Moderate (correlational)	2x risk	HHS Surgeon General, 2023
Technology use and well-being (individual level)	Strong (specification curve)	0.4% variance explained	Orben & Przybylski, 2019

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8. Bottom Line

The evidence on classroom technology's negative impacts has matured significantly. Dr. Jared Cooney Horvath's meta-synthesis of 398 meta-analyses provides the most comprehensive quantitative case yet: educational technology produces effects below the threshold for meaningful impact (d = 0.29) while consuming $30 billion annually in U.S. K-12 spending. International assessment data, experimental phone ban studies, and the documented "screen inferiority effect" for reading all point in the same direction — heavy classroom technology use is associated with worse academic outcomes, and reducing device access produces measurable gains.

However, the story is not a simple one of “technology bad, paper good.” Well-implemented educational technology does produce positive effects, particularly for disadvantaged students and those with specific learning needs. The equity implications of technology reduction are real: low-income students who rely on school as their primary technology access point could be further disadvantaged by broad pullbacks. And the workforce readiness argument — that 92% of jobs require digital skills — is not easily dismissed.

For a district like Greeneville City Schools, the practical path forward involves several steps:

1. **Compliance with HB 932 is a necessary first step**, and GCS has already adopted its cell phone policy. Early Tennessee results suggest this will produce measurable improvements in classroom behavior and engagement.

2. **The harder question is school-issued devices.** With 56% of educators nationally identifying school-issued device distraction as a bigger problem than cellphones, the cell phone ban alone does not address the primary source of classroom technology disruption. GCS should consider auditing how Chromebooks and other school-issued devices are actually being used — and how much instructional time is lost to off-task behavior.

3. **Software audit.** If GCS follows national patterns, a significant percentage of its software licenses are unused. A Denver-style consolidation could free resources for higher-impact investments.

4. **Intentional pedagogy, not blanket removal.** The evidence supports neither unrestricted technology use nor total elimination. The most effective approach is strategic: technology for adaptive feedback, assistive purposes, and cognitively active tasks; paper and direct instruction for reading, note-taking, and sustained attention.

5. **Monitor and measure.** Tennessee lacks state-level data linking 1:1 programs to TVAAS or TCAP outcomes. GCS could position itself as a leader by systematically tracking the relationship between technology use patterns and student growth data.

The questions that remain unanswered are significant: What is the optimal balance between digital and non-digital instruction? How should school-issued device policies evolve now that personal phone bans are in place? And will phone bans alone address the mental health concerns that motivated them, or are broader interventions needed? These are questions that no single study has resolved, and that individual districts will need to answer through careful, data-informed local practice.

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Appendix A: Tennessee Data

Tennessee TCAP Proficiency Rates

Subject	2019 (Pre-Pandemic)	2021-22	2022-23	2023-24	Trend
ELA (All Grades)	~33.7%	~34%	38.1%	39%	Exceeded pre-pandemic
Math (Grades 3-8)	Higher than current	~35%	~37%	40%	Still below 2019
Science (All)	~38%	~38%	~43%	44%	Record high
Social Studies (All)	~37%	~37%	~44%	45.3%	Record high

Note: Math recovery has been the slowest, which some national researchers have linked to heavy reliance on digital tools in math instruction during and after the pandemic. Tennessee has not produced a state-level study isolating technology as a variable.

Tennessee Legislative Actions on Technology in Schools

Bill	Year	Key Provision	Status
HB 932	2025	Classroom cell phone ban	Signed March 28, 2025
HB 1891	2024	Social media parental consent for minors	Signed May 2, 2024
HB 825	2025	Teen Social Media & Internet Safety Act	Signed; Phase 1 in effect
HB 1886	2026	School internet filtering & annual audits	In committee (passed 18-1)

Tennessee District Phone Ban Results

District	Enrollment	Method	Key Metric
Emerald Academy, Knoxville	Charter	Yondr pouches	80-90% of discipline was phone-related pre-ban
KIPP Nashville	Charter	Locking pouches	37% decrease in chronic absenteeism; 95% attendance

District	Enrollment	Method	Key Metric
East Nashville Magnet HS	~MNPS	Yondr pouches	83-87% of referrals were phone-related pre-ban; EOC pass rates improved
Greeneville City Schools	~2,500	Policy: off/out of sight	Implemented June 2025; monitoring in progress

Greeneville City Schools Cell Phone Policy (Adopted June 2025)

- Students must turn off or silence personal electronic devices during the school day
- Devices must be stored out of sight
- Smartwatches and headphones prohibited during school hours
- Exceptions for IEP/ILP requirements and assistive technology
- Discipline progression: parent notification, ISS (1-3 days escalating), loss of phone privileges for semester/year, OSS

Tennessee Context Factors

Factor	Data
National ranking in K-12 education funding	45th
National ranking in per-pupil spending	43rd
TISA technology line item	None (flexible allocation)
TVAAS studies linking 1:1 programs to growth	None published
TEAM evaluation technology indicators	None specific
RTI2 guidance on technology overuse	None specific
State-level ed-tech cost-effectiveness study	None conducted